Dynamical Systems Homework #1

Dmytro Zakharov

Problem (Exercise 1)**.** *Consider the following dynamical systems:*

(i) $X = \mathbb{R}^2/\mathbb{Z}^2$, $T(x,y) = (x+\alpha, y+\beta)$ *(for* $\alpha, \beta \in \mathbb{R}/\mathbb{Z}$*);*

(ii) $X = \mathbb{R}^2/\mathbb{Z}^2$, $T(x,y) = (ax, by)$ *(for* $a, b \in \mathbb{Z}$*).*

For each system, determine (depending on the parameters):

(a) Are there any fixed points?

(b) Are there any points with least period 2?

(c) Is the system minimal?

(d) Is the system transitive?

(e) Is the system topologically ergodic?

(f) Is the system topologically weakly mixing?

(g) Is the system topologically strongly mixing?

(h) Is the system equicontinuous?

Solution.

Part *(i)*.

Part (a). Fixed point $(x_0, y_0) \in \mathbb{R}^2/\mathbb{Z}^2$ must satisfy $x_0 + \alpha = x_0$ and $y_0 + \beta = y_0$. If $\alpha, \beta \in \mathbb{Z}$, then any point is a fixed point (in fact, the map T is such case is simply an identity map). Otherwise, T has no fixed points.

Further assume $\alpha, \beta \notin \mathbb{Z}$, otherwise the map T is trivial.

Part (b). Note than $T^n(x,y) = (x + n\alpha, y + n\beta)$. If the point (x_0, y_0) is a point with least period 2, then by definition $T^2(x_0, y_0) = (x_0, y_0)$ (note that the point cannot have a period 1 since we assumed $\alpha, \beta \notin \mathbb{Z}$), which translates to $2\alpha = 0$ and $2\beta = 0$. This means that $(\alpha, \beta) \in \{0, 1/2\}$. Otherwise, T has no points with least period 2.

Part (c). The system is minimal iff for any $(x,y) \in \mathbb{R}^2/\mathbb{Z}^2$ we have $X = \overline{\mathcal{O}(x,y)}$. Note that is suffices to consider when the orbit of point $(0,0)$ is dense (since for any other point $(a,b) \in \mathbb{R}^2/\mathbb{Z}^2$, the homeomorphism $\varphi_{a,b} : (x,y) \mapsto (x+a, y+b)$ maps the orbit of $(0,0)$ onto the orbit of (a,b), thus maps dense orbit to a dense orbit). The orbit of $(0,0)$ is precisely $\{(n\alpha, n\beta) : n \in \mathbb{Z}\}$. We claim that this set is dense iff $\{1, \alpha, \beta\}$ are linearly independent over $\mathbb{Q}$ (e.g., there is no $n, m \in \mathbb{Z} \setminus \{0\}$ such that $n\alpha + m\beta \in \mathbb{Z}$). Indeed, irrationality of both α and β is required since otherwise one of the components will not be dense. However, this condition is not sufficient (take, e.g., $\alpha := \sqrt{2}$ and $\beta := 2\sqrt{2}$): we need to ensure that we do not get the closed curve. This can only happen if $\alpha/\beta \in \mathbb{Q}$.

Part (d). If $\{1, \alpha, \beta\}$ are linearly independent over $\mathbb{Q}$, then as we have just shown, (X, T) is minimal and thus is transitive and topologically ergodic. If these elements were not linearly independent, then as we have shown, there would be no dense orbits, and thus the system cannot be transitive.

Part (e–g). Since topological ergodicity is equivalent to transitivity, the answer is the same as in *(d)*.

As we will see, the system is equicontinuous and thus is not topologically weakly mixing (and consequently not strongly mixing) for any choice of α, β.

Part (h). Recall that the definition of (X, T) being equicontinuous is as follows:

$$\forall_{(x_0,y_0)\in\mathbb{R}^2/\mathbb{Z}^2}\ \forall_{\varepsilon>0}\ \exists_{U\ni(x_0,y_0)}\ d(T^n(x_0,y_0),T^n(x_1,y_1))<\varepsilon\ \forall_{(x_1,y_1)\in U}\ \forall_{n\in\mathbb{Z}_{\geq 0}}$$

Since our space is a compact metric space,

$$\forall_{\varepsilon>0}\ \exists_{\delta>0}\ d((x_0,y_0),(x_1,y_1))<\delta \Rightarrow d(T^n(x_0,y_0),T^n(x_1,y_1))<\varepsilon\ \forall_{n\in\mathbb{Z}_{\geq 0}}$$

But note that since translation is a distance-preserving function, $d((x_0,y_0),(x_1,y_1)) = d(T^n(x_0,y_0),T^n(x_1,y_1))$, thus this condition is trivially satisfied (for $\delta := \varepsilon$). Thus (X, T) is equicontinuous for any values of α, β.

Part *(ii)*.

Part (a). Fixed point $(x_0, y_0) \in \mathbb{R}^2/\mathbb{Z}^2$ must satisfy $ax_0 = x_0$ and $by_0 = y_0$, which is equivalent to requiring $(a-1)x_0 \in \mathbb{Z}$ and $(b-1)y_0 \in \mathbb{Z}$. Note that for either $a = 1$ or $b = 1$, T acts as an identity to the corresponding component. Therefore, we have four cases:

(a) if $a = b = 1$, T is an identity, so all points are fixed;

(b) if $a = 1, b \neq 1$, then fixed points are $(\mathbb{R}/\mathbb{Z}) \times \{\frac{m}{b-1} : m \in \mathbb{Z}\}$; there are in total $|b-1|$ such fixed points;

(c) if $a \neq 1, b = 1$, then fixed points are $\{\frac{n}{a-1} : n \in \mathbb{Z}\} \times (\mathbb{R}/\mathbb{Z})$; there are in total $|a-1|$ such fixed points;

(d) if $a, b \neq 1$, then fixed points are $\{\frac{n}{a-1}, \frac{m}{b-1} : n, m \in \mathbb{Z}\}$; there are in total $|(a-1)(b-1)|$ such fixed points.

Part (b). Note that $T^n(x, y) = (a^n x, b^n y)$. The point (x_0, y_0) of minimal period 2 must satisfy $a^2x_0 = x_0$ and $b^2y_0 = y_0$, implying $(a^2-1)x_0 = (a+1)(a-1)x_0 \in \mathbb{Z}$ and $(b^2-1)y_0 = (b-1)(b+1)y_0 \in \mathbb{Z}$. However, these points cannot be fixed: in other words, such points are of form $(\frac{n}{a^2-1}, \frac{m}{b^2-1})$ where at least one of the conditions fail: $(a+1) \mid n$ or $(b+1) \mid m$.

We also need to handle special cases when $a = \pm 1$ and $b = \pm 1$. Notice that when $a = 1$, no point can have period 2 in x-coordinate, while for $a = -1$, every point has period 2 in x-coordinate. Same holds for b.

Part (c). The system is never minimal. Indeed, if either a or b is not 1, then from *part (a)* we know that the system has fixed points. If $a = b = 1$, then T is the identity map which clearly implies the system is not minimal either.

Part (d). We need to understand when the orbit's closure $\overline{\mathcal{O}(x_0, y_0)}$ of some $(x_0, y_0) \in \mathbb{R}^2/\mathbb{Z}^2$ is dense. Clearly, if $|a| = |b| = 1$, all orbits are finite and thus not dense, so the system is not transitive. However, when both $|a|, |b| \geq 2$, then notice that any open set U after applying T stretches and eventually intersects any open set V (thus making $T^n(U) \cap V \neq \emptyset$). If either of $|a|$ or $|b|$ is 1, then one of the coordinates is periodic, and the orbit stays inside closed circles. Thus, the closure is not dense.

Conclusion: (X, T) is transitive iff $|a| \geq 2$ and $|b| \geq 2$.

Part (e). Equivalent to *part (d).*

Part (f). The system is weakly mixing whenever $(X \times X, T \times T)$ is topologically ergodic, or simply transitive in our case. So in other words, we are given the system $\mathbb{R}^4/\mathbb{Z}^4$ with the map $T : (x, y, z, w) \mapsto (\alpha x, \beta y, \alpha z, \beta w)$ and we need to decide

whether this system has a transitive point. However, this is clearly the case as long as $|a|, |b| \geq 2$ (with the similar reasoning to *part (d)*). So the conclusion is equivalent to *part (d)*.

Part (g). The system is similarly strongly mixing whenever $|a|, |b| \geq 2$.

Part (h). Recall that for (X, T) to be equicontinuous, one has to have

$$\forall_{\varepsilon>0}\ \exists_{\delta>0}\ d((x_0, y_0), (x_1, y_1)) < \delta \Rightarrow d((a^n x_0, b^n y_0), (a^n x_1, b^n y_1)) < \varepsilon\ \forall_{n\in\mathbb{Z}_{\geq 0}}$$

However, notice that if either $|a| \geq 2$ or $|b| \geq 2$, then the distance between two points (in the corresponding coordinate) exponentially increases (modulo 1). For instance, consider two very close points (x_0, y_0) and $(x_0 + \delta, y_0)$ for some $\delta \ll 1$. After n iterations, the distance in the first distance becomes $|a|^n \delta$, which grows exponentially (albeit wrapped around the circle).

When both $|a| = |b| = 1$, then the map T is distance-preserving, thus the system is equicontinuous.

Conclusion. (X, T) is equicontinuous iff $|a| = |b| = 1$.

Problem (Exercise 2). *Let (X, T) be a topological dynamical system, and assume that X is compact and metrizable, with a metric d.*

(a) Let d' be another metric on X. Show that (X, T) is equicontinuous with respect to d iff (X, T) is equicontinuous with respect to d'.

(b) Show that (X, T) is equicontinuous iff the following condition is satisfied:

$$\forall_{\varepsilon>0}\ \exists_{K\subseteq\mathbb{N}, K \text{ finite}}\ \forall_{n\in\mathbb{N}}\ \exists_{k\in K}\ d(T^n(x), T^k(x)) < \varepsilon\ \forall_{x\in X}.$$

Solution.

Part *(a)*. Recall that on a compact space, two compatible metrics are uniformly equivalent, i.e.,

$$(1) : \forall_{\varepsilon>0}\ \exists_{\delta>0}\ \{d'(x,y) < \delta\} \Rightarrow \{d(x,y) < \varepsilon\}.$$

Fix $\varepsilon > 0$ and choose $\delta := \varepsilon'$ that satisfies the above inequality. By equicontinuity in metric d, there exists $\delta > 0$ such that

$$(2) : \{d(x,y) < \delta\} \Rightarrow \{d(T^n(x), T^n(y)) < \varepsilon'\}\ \forall_{n\in\mathbb{N}}.$$

Again by uniform equivalence, choose $\delta' > 0$ such that

$$(3) : \{d'(x,y) < \delta'\} \Rightarrow \{d(x,y) < \delta\}.$$

Then, for any $x, y \in X$, we have

$$\begin{aligned}\{d'(x,y) < \delta'\} &\overset{(3)}{\Longrightarrow} \{d(x,y) < \delta\} \\ &\overset{(2)}{\Longrightarrow} \{d(T^n(x), T^n(y)) < \varepsilon'\}\ \forall_{n\in\mathbb{N}} \\ &\overset{(1)}{\Longrightarrow} \{d'(T^n(x), T^n(y)) < \varepsilon\}\ \forall_{n\in\mathbb{N}}.\end{aligned}$$

Part *(b)*. We prove two directions.

Equicontinuous $\Rightarrow$ finite covering in d_∞. Fix some $\varepsilon > 0$. By equicontinuity, for each $x \in X$ there exists $\delta > 0$ such that

$$(1) : \{d(x,y) < \delta\} \Rightarrow \{d(T^n(x), T^n(y)) < \varepsilon/3\}\ \forall_{n\in\mathbb{N}}.$$

Since X is compact, we can cover X by finitely many balls of radius δ: concretely, let $X = \bigcup_{i=1}^{N} B(x_i, \delta)$. However, these centers on their own do not help much. We need to understand the dynamics of the sequence $\{(T^n(x_1), \ldots, T^n(x_i))\}_{n\in\mathbb{N}}$ which lies in the space X^N (which further denote by $\{\mathbf{x}_n\}_{n\in\mathbb{N}}$).

That said, we consider the space (X^N, d_∞) which we equip with the metric $d_\infty(\mathbf{x}, \mathbf{y}) = \max_{1\le i\le N} d(x_i, y_i)$. Clearly, X^N is compact (say, by Tychonoff's Theorem), so the sequence $\{\mathbf{x}_n\}_{n\in\mathbb{N}}$ is totally bounded. In particular this means that we can find a finite set of indices $K \subseteq \mathbb{N}$ such that $\{\mathbf{x}_k\}_{k\in K}$ is the $(\varepsilon/3)$-covering of our sequence $\{\mathbf{x}_n\}_{n\in\mathbb{N}}$ in d_∞ metric. Specifically:

$$\forall_{n\in\mathbb{N}}\ \exists_{k\in K}\ \{d_\infty(\mathbf{x}_n, \mathbf{x}_k) < \varepsilon/3\}.$$

So by expanding everything we have so far:

$$(2) : \exists_{K\subseteq\mathbb{N}, K \text{ is finite}}\ \forall_{n\in\mathbb{N}}\ \exists_{k\in K}\ \{d(T^n(x_i), T^k(x_i)) < \varepsilon/3\} \text{ for each } i \in \{1,\dots,N\}.$$

Now, we have all the tools to finish off the statement. Fix $\varepsilon > 0$. Take K as in (2). For every $x \in X$ there exists some center of the ball x_i such that $d(x, x_i) < \delta$. For every $n \in \mathbb{N}$ choose k as in (2). Then, apply the triangle inequality:

$$d(T^n(x), T^k(x)) \leq \underbrace{d(T^n(x), T^n(x_i))}_{<\varepsilon/3 \text{ by } (1)} + \underbrace{d(T^n(x_i), T^k(x_i))}_{<\varepsilon/3 \text{ by } (2)} + \underbrace{d(T^k(x_i), T^k(x))}_{<\varepsilon/3 \text{ by } (1)} < \varepsilon.$$

Finite covering in $d_\infty \Rightarrow$ Equicontinuous. Assume

$$\forall_{\varepsilon>0}\ \exists_{K\subseteq\mathbb{N}, K \text{ finite}}\ \forall_{n\in\mathbb{N}}\ \exists_{k\in K}\ \forall_{x\in X}\ d(T^n(x), T^k(x)) < \varepsilon.$$

Fix $\varepsilon > 0$ and pick any two points $x, y \in X$.

From the fact that T^k is uniformly continuous (in particular for all $k \in K$ since X is compact), we find $\delta > 0$ such that:

$$\{d(x,y) < \delta\} \Rightarrow \forall_{k\in K}\ \{d(T^k(x), T^k(y)) < \varepsilon/3\}.$$

Then, for any n, pick $k \in K$ with $d(T^n(x), T^k(x)) < \varepsilon/3$ and $d(T^n(y), T^k(y)) < \varepsilon/3$. Finally, apply the triangle inequality:

$$d(T^n(x), T^n(y)) \leq d(T^n(x), T^k(x)) + d(T^k(x), T^k(y)) + d(T^k(y), T^n(y)) < \varepsilon,$$

hence (X, T) is equicontinuous.

Problem (Exercise 3). *Consider the following maps on* $\mathbb{R}$*:*

$$T_1(x) = x,\ T_2(x) = -x,\ T_3(x) = x + 3,$$
$$T_4(x) = 4x,\ T_5(x) = 4x(1-x),\ T_6(x) = 6x(1-x).$$

Show that no two of the resulting topological dynamical systems (X, T_i) $(1 \leq i \leq 6)$ *are isomorphic.*

Solution.

Idea. To prove that these topological dynamical systems are distinct, we need to find distinguishing invariant of each system. In this problem it suffices to consider periodic points and invariant sets. Formally, the following proposition states that these two are indeed invariants of the dynamical system under isomorphism.

Lemma 0.1 (Adapted Proposition 1.5.2, [Vri14]). *Let* $\varphi : (X, T) \xrightarrow{\sim} (Y, S)$ *be the isomorphism of dynamical systems. Then,*

(a) point x *is periodic in* (X, T) *iff* $\varphi(x)$ *is periodic in* (Y, S) *(moreover, primitive periods of both* x *and* $\varphi(x)$ *are the same);*

(b) set A *in* (X, T) *is* T*-invariant iff* $\varphi(A)$ *is* S*-invariant in* (Y, S)*.*

Proof. By considering periodic points, we can differentiate systems T_1, T_2, T_3, T_4, and (T_5 *with* T_6) by applying Lemma, *part (a)*. To differentiate T_5 and T_6, we are going to look at the invariant sets and apply Lemma, *part (b)*, although this part turned out to be the most challenging in this problem.

System T_1. This is just an identity map, so it is clearly distinct from other systems: for instance, since every point is fixed.

System T_2. Note that the only fixed points on this map is $x = 0$ and all other points have the primitive period of 2. No other systems have this characterization.

System T_3. This system, in contrast, has no periodic points (all other systems, in turn, have at least a single periodic point).

Systems T_4. This systems has a fixed point $x = 0$ and no other periodic points.

Systems T_5 *and* T_6. This one is the hardest. Let us consider the general case $f_\mu(x) = \mu x(1-x)$ (the so-called *logistic map*). By solving equation $f_\mu(x) = x$ we get that it always (besides the case $\mu = 0$) has two solutions: $x_0 = 0$ and $x_1 = 1 - 1/\mu$. Although this already differentiates T_5 and T_6 from the rest of the systems, we need another measure to differentiate T_5 and T_6.

We propose to compute the maximum value of f_μ on the set $[0, 1] \subseteq X$. Notice that $f'_\mu(x) = \mu(1 - 2x)$, so the extremum point corresponds to the value at $x_{\text{ext}} = \frac{1}{2}$ (which lies within $[0, 1]$). Moreover, $f_\mu(x_{\text{ext}}) = \frac{1}{4}\mu$. Thus, we've shown the following:

Claim #1. The set $[0, 1]$ is f_μ-invariant iff $0 \leq \mu \leq 4$.

Therefore, the dynamical system T_5 has an invariant set with non-empty interior. Consequently, by Lemma, *part (b)*, if there were the isomorphism between (X, T_5) and (X, T_6), it would correspond to the invariant set with non-empty interior in (X, T_6). However, our second claim is as follows.

Claim #2. The dynamical system (X, f_μ) has no invariant sets with non-empty

interior for $\mu > 2 + \sqrt{5}$.

Proof of claim. First we observe that the orbit of any point outside $[0, 1]$ "explode" to $-\infty$ (for instance, take any $x_0 > 1$, then $x_1 := f_\mu(x_0)$ becomes negative, and the next iteration will scale x_1 by the factor $\mu(1 - x_1)$, which is clearly more than 1, and thus will repel the point further to $-\infty$).

Therefore, if we had an invariant set $A \subseteq X$, then the orbit $\mathcal{O}(x)$ of any $x \in A$ must remain in $[0, 1]$. Let us check whether this holds.

Let us check the "escape region" $E \subseteq X$, from which we get values greater than 1: i.e., $f_\mu(x) > 1$. This yields equation $\mu x^2 - \mu x + 1 < 0$ which gives the interval $E = (\ell, r)$ for

$$\ell = \frac{1}{2} - \frac{1}{2}\sqrt{1 - \frac{4}{\mu}}, \quad r = \frac{1}{2} + \frac{1}{2}\sqrt{1 - \frac{4}{\mu}}.$$

(Note that this is a valid non-empty interval as $\mu > 4$).

Therefore, the "safe regions" of f_μ are two zones: $L = [0, \ell]$ and $R = [r, 1]$: if the point happens to be in (ℓ, r), it will be sent to infinity.

Now take any open interval $(a, b) \subseteq A$. Without loss of generality, assume $(a, b) \subseteq L$ or $(a, b) \subseteq R$ (otherwise, if $E \cap (a, b) \neq \emptyset$, then clearly A is not invariant). We claim that $f_\mu^n((a, b))$ eventually intersects E. Indeed, it suffices to verify that this interval gets expanded with each iteration of f_μ. To check this, we check the minimum value of $|f'_\mu|$ in the "safe" zone $L \cup R$*.

Note that since $f'_\mu(x)$ is a linear function, its magnitude strictly decreases at x gets closer to the critical point $x_{\text{ext}} = \frac{1}{2}$. Thus, the minimum value of $|f'_\mu(x)|$ on $L \cup R$ occurs at points ℓ and r. One easily checks $|f'_\mu(\ell)| = |f'_\mu(r)| = \sqrt{\mu^2 - 4\mu}$. If this value exceeds 1, then the claim is proven. This occurs when $\mu^2 - 4\mu - 1 > 0$ which reduces to $\mu > 2 + \sqrt{5}$.

Corollary. (X, T_6) has no invariant sets with non-empty interior.

Answer. See Table 1 below.

System	Periodic Points	Invariant Sets A (**int**$(A) \neq \emptyset$)
T_1	Every point is a fixed point	N/A
T_2	$x = 0$ is fixed; others have min. period 2	N/A
T_3	None	N/A
T_4	$x = 0$ is fixed; no others	N/A
T_5	Two fixed points ($x = 0, x = 3/4$)	Yes ($[0, 1]$ is invariant)
T_6	Two fixed points ($x = 0, x = 5/6$)	None

Table 1: Comparison of distinguishing invariants for dynamical systems (X, T_i).

*Recall that from Mean Value Theorem, $f(x_2) - f(x_1) = f'(c)(x_2 - x_1)$ for some $c \in [x_1, x_2]$.

Problem (Exercise 4). *We call a topological dynamical system (X, T) totally minimal (resp. totally transitive) if (X, T^d) is minimal (resp. transitive) for all $d \geq 1$.*

(a) Show that for $\alpha \in \mathbb{R} \setminus \mathbb{Q}$, the circle rotation by α is totally minimal.

(b) Show that the doubling map on the circle is totally transitive.

(c) Show that not each minimal (resp. transitive) topological dynamical system is totally minimal (resp. totally transitive).

Solution.

Part *(a)*. It is well-known that the map $R_\alpha : x \mapsto x + \alpha$ (over $\mathbb{R}/\mathbb{Z}$) is minimal iff $\alpha \in \mathbb{R} \setminus \mathbb{Q}$. Note that $R_\alpha^d = R_{d\alpha}$ and $d\alpha$ is clearly irrational as well, thus R_α^d is also minimal. Hence, (X, R_α) is totally minimal.

Part *(b)*. We already know well that the doubling map is transitive. In fact, any map $M_\gamma : x \mapsto \gamma x$ for integer $\gamma \geq 2$ is transitive. The argument is similar to the case when $\gamma = 2$: we need to show that for any opens $U, V \subseteq \mathbb{R}/\mathbb{Z}$ there exists $n \in \mathbb{N}$ such that $T_\gamma^n(U) \cap V \neq \varnothing$. If U is an open interval of length $\delta > 0$, then after n iterations the "length" of $T_\gamma^n(U)$ is $\gamma^n \delta$. For large enough n this exceeds 1 and so $T_\gamma^n(U)$ effectively wraps the original U around the circle at least once. Thus, as we cover the whole circle, we definitely intersect V at some point.

That said, the nth iterate of the doubling map satisfies $M_2^n = M_{2^n}$ which is transitive by the argument above for every n. Thus, (X, M_2) is totally transitive.

Part *(c)*.

Minimal but not totally minimal. Let $X = (\mathbb{R}/\mathbb{Z}) \times \mathbb{Z}_2$ (where $\mathbb{Z}_2$ is equipped with discrete topology) and $T : (x, b) \mapsto (x + \alpha, b + 1)$ (with $\alpha \in \mathbb{R}/\mathbb{Z}$). Clearly, T is minimal, but is not totally minimal since $T^2(x, b) = (x + 2\alpha, b)$ is not minimal as it fixes the second coordinate (thus, say, $(\mathbb{R}/\mathbb{Z}) \times \{0\}$ is a smaller T^2-invariant set).

Transitive but not totally transitive. Similarly consider $X = (\mathbb{R}/\mathbb{Z}) \times \mathbb{Z}_2$ and $T : (x, b) \mapsto (2x, b + 1)$. Then, T is transitive while $T^2(x, b) = (4x, b)$ is not since the only possible orbits are $(\mathbb{R}/\mathbb{Z}) \times \{0\}$ or $(\mathbb{R}/\mathbb{Z}) \times \{1\}$.

Problem (Exercise 5). *Let $\phi : X \to Y$ be a morphism between topological dynamical systems (X, T) and (Y, S). Assume that X is locally compact, and let $A \subseteq X$ be non-empty, compact and invariant.*

(a) Show that $\phi(\mathcal{B}_T(A)) \subseteq \mathcal{B}_S(\phi(A))$, where $\mathcal{B}_T(A)$ denotes the basin of attraction of the set A under the map T.

(b) Show that the inclusion in the previous point can be strict.

Solution.

Part *(a)*. Recall that the basin of attraction is by definition

$$\mathcal{B}_T(A) = \{x \in X : \emptyset \neq \omega_T(x) \subseteq A\}.$$

We prove this fact by the series of three claims.

Claim #1 Assume X is locally compact. If $x \in X$ and $\omega(x)$ is compact and not empty, then $\overline{\mathcal{O}}(x)$ is compact.

Claim #2 If $\overline{\mathcal{O}}(x)$ is compact then $\phi(\omega_T(x)) = \omega_S(\phi(x))$.

Claim #3 Assume $\phi(A)$ is a closed subset of Y and all $\overline{\mathcal{O}}(x)$ for $x \in \mathcal{B}_T(A)$ are compact. Then, $\phi(\mathcal{B}_T(A)) \subseteq \mathcal{B}_S(\phi(A))$.

Proof of claims.

Claim #1. Since X is locally compact, $\omega(x)$ has an open neighbourhood U with compact closure $\overline{U}$. Note that $T^n(x) \in U \subseteq \overline{U}$ holds for infinitely many values of n. Thus, it suffices to show that $T^n(x) \in \overline{U}$ for almost all values of n. Assume, however, that it is not the case. Then, there is a sequence $\{T^{n_i}(x)\}_{i\in\mathbb{N}} \subseteq \overline{U}$ such that $T^{n_i+1}(x) \notin \overline{U}$ for all i. The first sequence has an accumulation point z in $\omega(x)$, but the second sequence has an accumulation point $T(z)$ outside U, contradicting invariance of $\omega(x)$.

Claim #2. Note that in general we have $\phi(\omega_T(x)) \subseteq \omega_S(\phi(x))$:

$$\begin{aligned}
\phi(\omega_T(x)) &\triangleq \phi\left(\bigcap_{n\geq 0} \overline{\mathcal{O}}_T(T^n(x))\right) && \text{// definition of limit set}\\
&\subseteq \bigcap_{n\geq 0} \phi(\overline{\mathcal{O}}_T(T^n(x))) && \text{// use the fact that } \phi \text{ is continuous}\\
&\subseteq \bigcap_{n\geq 0} \overline{\phi(\mathcal{O}_T(T^n(x)))} && \text{// again, since } \phi \text{ is continuous}\\
&= \bigcap_{n\geq 0} \overline{\mathcal{O}}_S(\phi(T^n(x))) && \text{// just a direct observation}\\
&= \bigcap_{n\geq 0} \overline{\mathcal{O}}_S(S^n(\phi(x))) && \text{// definition of morphism}\\
&= \omega_S(\phi(x)) && \text{// definition of limit set}
\end{aligned}$$

Note that we remove inclusions at second and third step using the fact that $\overline{\mathcal{O}}(x)$ is compact.

Claim #3. By *Claim #2*, $\omega_S(\phi(x)) = \phi(\omega_T(x))$ is a non-empty subset of $\phi(A)$ for all $x \in \mathcal{B}_T(A)$, hence $\phi(\mathcal{B}_T(A)) \subseteq \mathcal{B}_S(\phi(A))$.

Now simply apply three claims. Since A is compact, then for every point $x \in \mathcal{B}_T(A)$, the set $\omega(x)$ is non-empty and compact. By *Claim #1*, $\overline{\mathcal{O}}(x)$ is compact, so by *Claim #3* we have proven the theorem.

Part *(b)*. Take the following example:

- System (X, T). Let $X = \{0\} \cup [2, 3]$ and let

$$T(x) = \begin{cases} 0, & x = 0, \\ 1 + \frac{1}{2}x, & x \in [2, 3]. \end{cases}$$

  Set $A = \{0\}$ (so that $\mathcal{B}_T(A) = \{0\}$).
- System (Y, S). Let $Y = [0, 1]$ and $S(y) = y/2$.

Let the morphism $\phi : (X, T) \to (Y, S)$ be

$$\phi(x) = \begin{cases} 0, & x = 0, \\ x - 2, & x \in [2, 3]. \end{cases}$$

This is a valid morphism: it is continuous and satisfies $\phi(T(x)) = S(\phi(x))$.

Finally, note:

- $\phi(\mathcal{B}_T(A)) = \phi(\{0\}) = \{0\}$.
- $\mathcal{B}_S(\phi(A)) = \mathcal{B}_S(\{0\}) = [0, 1]$ as any point of Y converges to 0.

Thus, we see that $\{0\} = \phi(\mathcal{B}_T(A)) \subsetneq \mathcal{B}_S(\phi(A)) = [0, 1]$.

References

[Vri14] J. Vries. *Topological Dynamical Systems: An Introduction to the Dynamics of Continuous Mappings*. De Gruyter Studies in Mathematics. De Gruyter, 2014. URL: https://books.google.com.ua/books?id=hyroBQAAQBAJ.